Equivalent ratios

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: I can keep a ratio unchanged while making more or less of a mixture.

R1. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

R2. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

R3. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

R4. Find $\frac{3}{4} \times \frac{4}{5}$.

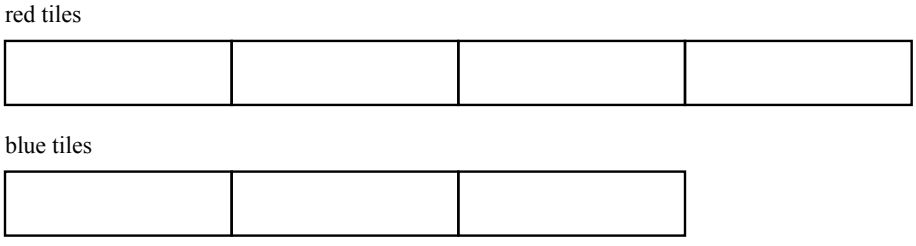
P1. There are 4 red tiles and 3 blue tiles. Write red : blue. Write blue : red.

P2. A trail mix uses 4 raisins for every 3 nuts. How many nuts go with 12 raisins?

P3. A box has 20 red tiles and 15 blue tiles. Divide both counts by 5 to find an equivalent ratio.

P4. A recipe uses 6 cups of oats for every 4 cups of seeds. How many cups of seeds go with 9 cups of oats?

Lesson model



Equivalent ratios

Practice and compare methods

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P5. Paint uses 2 cups yellow for every 5 cups blue. How much yellow goes with 15 cups blue?

P6. Each game team has 3 builders for every 2 testers. How many builders go with 8 testers?

F. A mix uses 5 cups oats for every 2 cups seeds. Ten cups oats makes __ batches and needs __ cups seeds.

D. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

X1. Red: blue is 2 : 7. Complete (2, 7), (4,__).

X2. A mix has 15 raisins and 35 nuts. Divide both counts by 5; write the new ratio.

X3. A recipe uses 4 cups oats for 10 cups seeds. How many cups seeds go with 6 cups oats?

X4. Is 8 : 10 equivalent to 3 : 5? Explain without using the difference.

Worked example - compare with the slide model

A mix uses 3 cups oats for every 2 cups seeds. Find seeds for 12 cups oats.

12 ÷ 3 = 4 batches.

Each batch has 2 cups seeds.

4 × 2 = 8 cups seeds.

Result: 8 cups seeds

Equivalent ratios

Challenge, explain and check

20 daily questions include C1 and C2. Use separate paper when helpful.

C1. A mosaic uses 3 red tiles for every 5 blue tiles. You use 18 red tiles and have 32 blue tiles. How many blue tiles are needed? How many remain?

C2. A trail mix uses 4 cups oats for 6 cups seeds. You have 10 cups oats and 18 cups seeds. Use all the oats in the same ratio. How many cups seeds remain?

Finish the lesson

S. To keep a ratio the same, I ___ both quantities by ___.

E. A bracelet pattern uses 7 green beads for every 4 white beads. How many green beads go with 12 white beads?